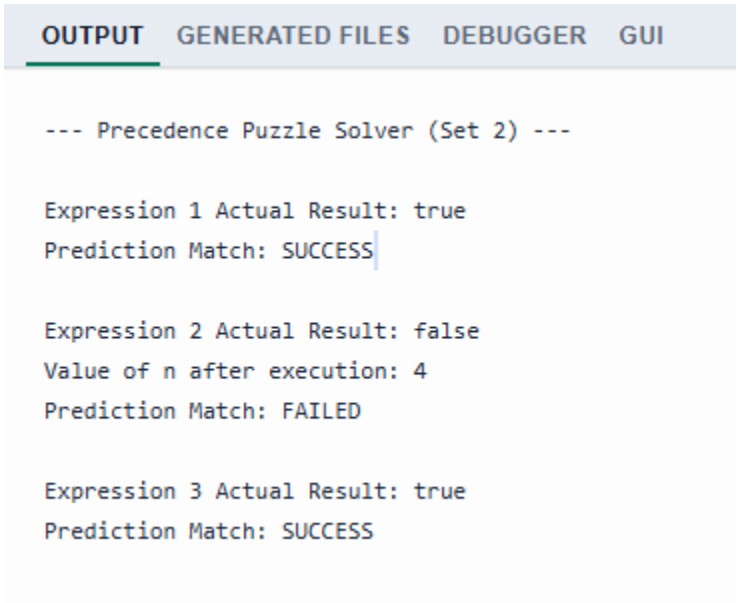
```
public class PrecedencePuzzleSolver {
    public static void main(String[] args) {
        System.out.println("--- Precedence Puzzle Solver ---\n");

        // EXPRESSION 1: Combining Post-Increment, Division, and Relational Operators
        int m = 10;

        // PREDICTED RESULT: true
        // Reasoning: m++ evaluates to 10 first (then m becomes 11 in memory).
        // 10 / 2 resolves to 5. Finally, 5 >= 5 evaluates to true.

        boolean result1 = m++ / 2 >= 5;
        System.out.println("Expression 1 Actual Result: " + result1);
        System.out.println("Prediction Match: SUCCESS\n");



        // EXPRESSION 2: Logical AND Short-Circuiting (Intentional Mistake)
        int n = 4;

        // PREDICTED RESULT: The variable 'n' will equal 5 after this line runs.
        // (Mistakenly assuming that both sides of an '&&' operator always evaluate)

        boolean result2 = (n > 10) && (++n == 5);
        System.out.println("Expression 2 Actual Result: " + result2);
        System.out.println("Value of n after execution: " + n);
        System.out.println("Prediction Match: FAILED");

        /*
         * FOLLOW-UP COMMENT (Precedence / Evaluation Rule Missed):
         * The prediction turned out wrong because of Java's Short-Circuit Evaluation rule.
         * For the logical AND (&&) operator, if the left side evaluates to false (4 > 10 is false),
         * Java completely skips evaluating the right side because the entire statement can never
be true.
         * As a result, the right side (++n == 5) was never executed, and 'n' remained 4 instead of
becoming 5.
         */
        System.out.println();



        // EXPRESSION 3: Mixed Arithmetic, Equality, and Inequality Operators
        int k = 6;

        // PREDICTED RESULT: true
```

```
        // Reasoning: Arithmetic happens first: k - 1 is 5.
        // Relational happens next: k > 10 (6 > 10) is false.
        // Equality runs next: 5 == 5 is true.
        // Finally, the inequality operator compares them: true != false resolves to true.

        boolean result3 = k - 1 == 5 != (k > 10);
        System.out.println("Expression 3 Actual Result: " + result3);
        System.out.println("Prediction Match: SUCCESS\n");
    }
}
```